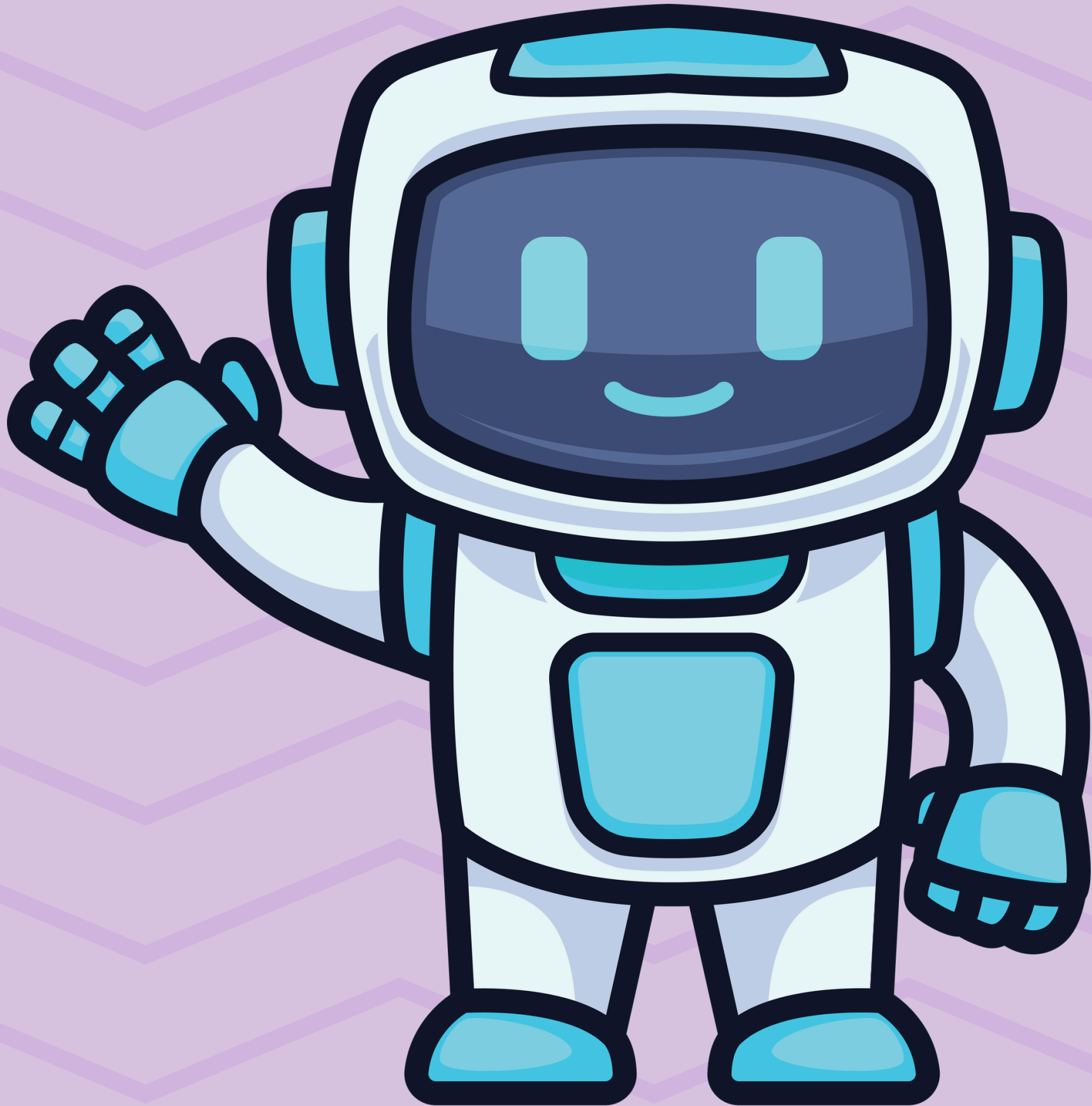


ESCAPE ROOM

GAME

ISC COMPUTER
SCIENCE LAB 2026



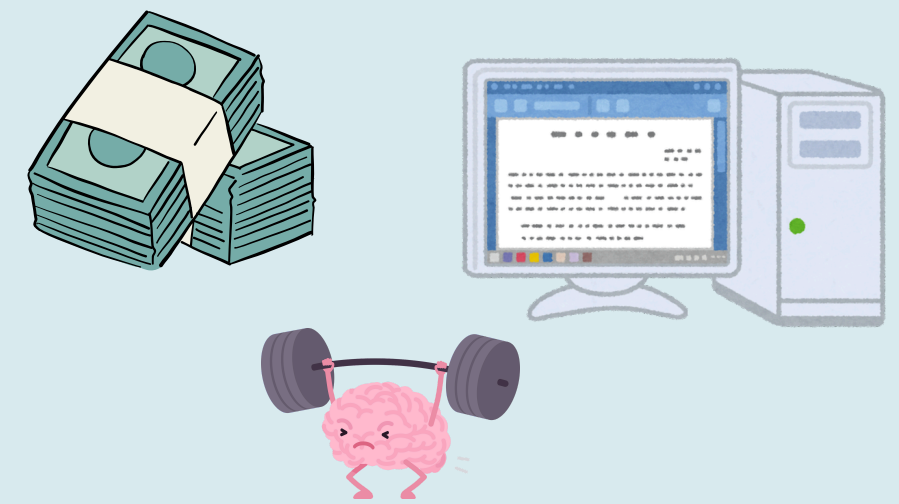


BUT FIRST, WHAT IS COMPUTER SCIENCE?

Computer science is the study of computers, computational systems, and algorithmic processes, covering both theoretical foundations (algorithms, data structures) and practical applications (software/hardware design, AI).

It focuses on problem-solving, using mathematics and logic to create, test, and improve software systems to solve real-world challenges.

WHERE CAN COMPUTER SCIENCE GET YOU?



- Software Development & Engineering: Building applications, websites (front-end/back-end), and mobile apps.
- Data Science & AI: Analyzing complex datasets, creating machine learning models, and designing AI systems.
- Cybersecurity & Infrastructure: Securing networks, acting as IT consultants, or working as systems administrators.
- Specialized Tech: Working with blockchain, Internet of Things (IoT), virtual reality (AR/VR), or robotics.
- Management & Strategy: Roles such as Product Manager, Chief Technology Officer (CTO), or IT Project Manager.

SETUP

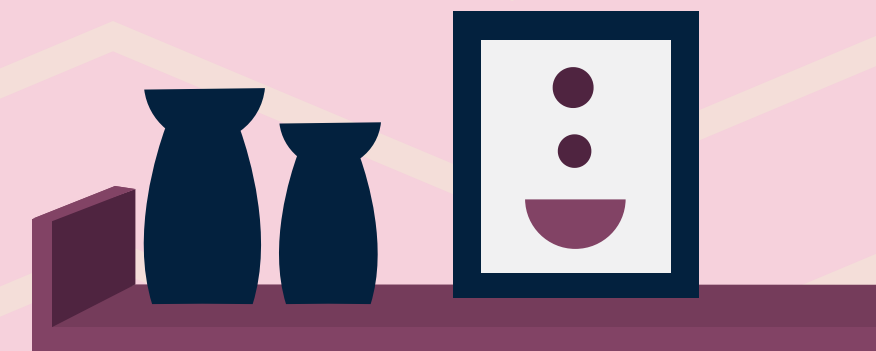
You are tasked with coding an escape room of three puzzles in python.

Lab Assignment: 55 marks out of 100, with a 5 mark bonus problem!

Github repository:

Question Package:

- Print copies will be distributed to students to fill out.
- A total of 45 marks out of 100 for the lab.
- Hand-written answers only.



WHAT IS A FUNCTION?

Definition:

A function is a named block of code that:

- performs a specific task
- can be reused multiple times

Why we use functions:

- Avoid repeating code
- Make programs easier to read
- Break big problems into smaller steps

Think of a function like a machine:

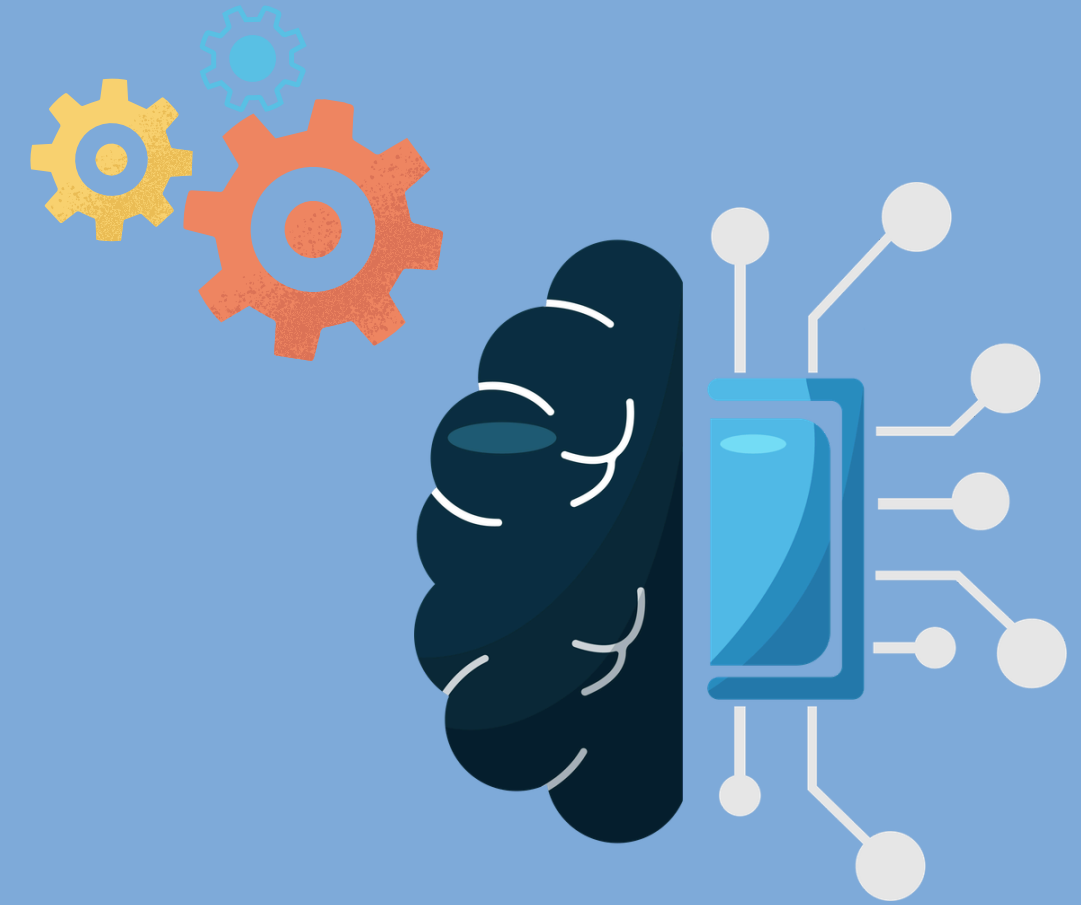
input → work → output



FUNCTION DEFINITION

python

```
def function_name(parameters):  
    # code goes here  
    return value
```

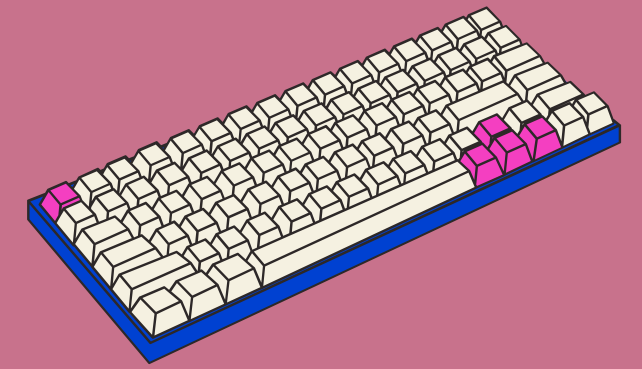


Important parts

- `def` → tells Python you are creating a function
- `function_name` → what you call it
- `parameters` → inputs (optional)
- `return` → sends a value back (optional)

Not all functions need parameters or a return value.

EXAMPLE



```
def calculate_total(price, quantity):  
    total = price * quantity  
    return total
```

What this function does:

- Takes a price and quantity
- Multiplies them
- Returns the total cost

Calling the Function

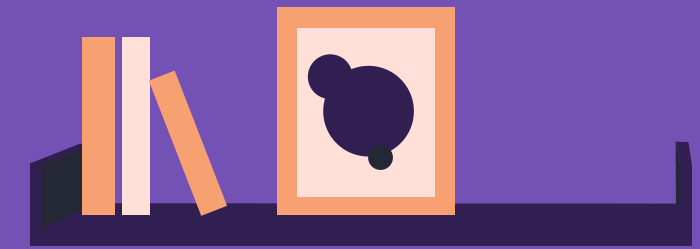
```
cost = calculate_total(4.50, 3)  
print(cost)
```

What happens

1. `calculate_total(4.50, 3)` runs
2. The function returns 13.5
3. The value is stored in `cost`
4. `print(cost)` displays the result

PUZZLE 1

- You'll navigate a room by entering (x, y) coordinates on a grid.
- Your job is to write code that checks whether a coordinate is inside the grid and not on a laser.
- Have a goal state you need to reach with moves.
- If you step on a laser or go out of bounds, you'll need to try again.



C A E S A R

— — — — —

PUZZLE 2

- You'll find a secret message where each letter has been shifted forward by 3 in the alphabet.
- Write a function to decode the message by shifting letters back.
- Once you find the hidden key phrase, the lock will open.

PUZZLE 3

- The final door uses a Fibonacci sequence as its security code.
- You'll generate the first n Fibonacci numbers using a loop.
- The last number in the sequence unlocks the door.





FINAL STEP

Each puzzle needs to be passed to escape the room!



THINGS TO REMEMBER

- Use while loops when the program should keep asking until the input is correct.
- Handle bad input gracefully (non-numbers, out-of-range values, etc.) so your program doesn't crash.
- Test your functions often. Small print statements and throwing errors will help you debug faster.
- Read the function comments carefully, they tell you what to return, not just what to print.



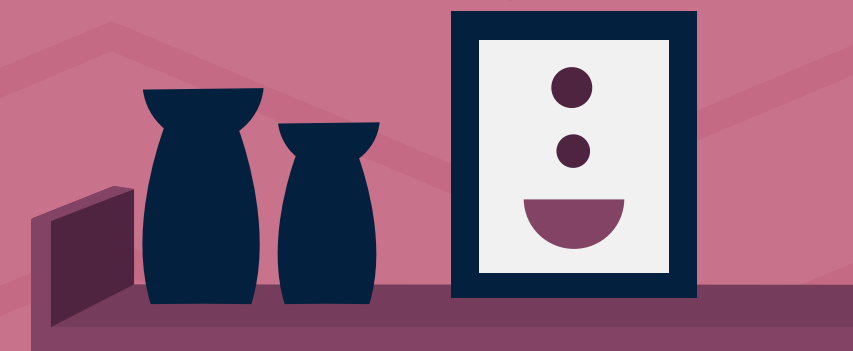
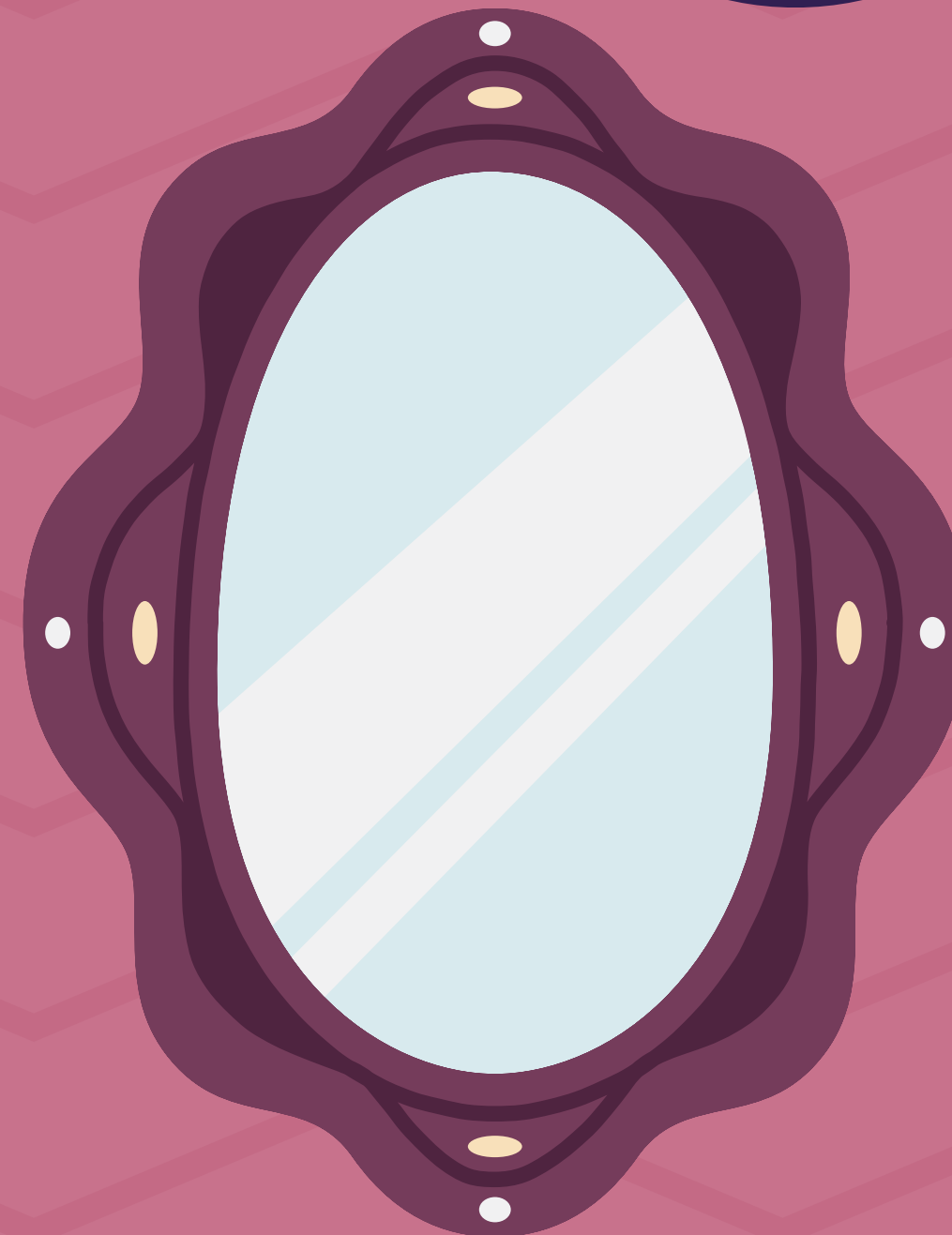
LASTLY...

You should be able to complete both the Question Package and the main lab component in our 3 hour block.

Bonus marks are only an additional 5 points so leave it to the end
There is nothing wrong with asking for help, look around for the TAs around you.

Coding is hard. But we are here to have fun! Showing your effort will still earn you points!

Any use of AI is strictly prohibited!



GOOD LUCK!

